

Autodesk® Scaleform®

APIs

Scaleform 4.4

Scaleform C++

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2.2

2012 6 21

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Autodesk® Scaleform® 4.4

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1 Scaleform

Scaleform HTML GfX/SWF
FreeType 2

Scaleform Flash® Studio SWF
Flash Studio
Flash

Scaleform GfX::Translator GfX::FontLib
FreeType2
Flash Scaleform Flash Studio Scaleform
Flash
Scaleform
Flash Scaleform

- “Flash” Flash
Flash
- “” Flash

1.1 Flash

Flash Studio

Studio

Flash

Scaleform

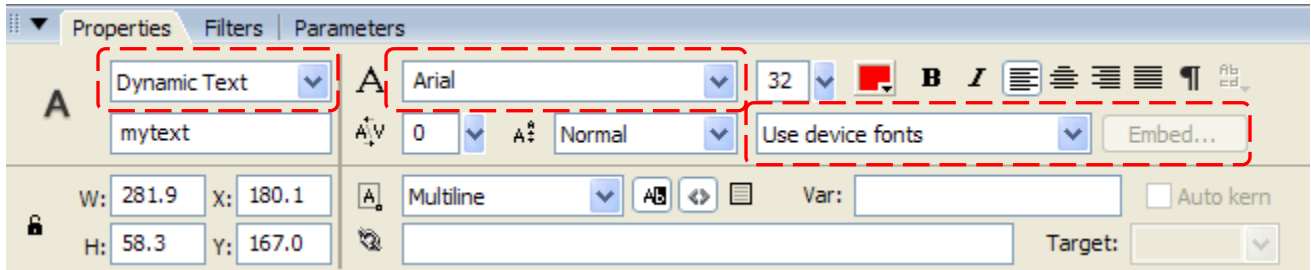
1 ---

1.1.1

Flash

Text Tool

Flash Studio



-
-
-

GFx::Translator

(a)

“Arial”

(b)

SWF

Flash 8

-
-

-
-

“Use device fonts” “ ”

SWF
Flash
/

Scaleform

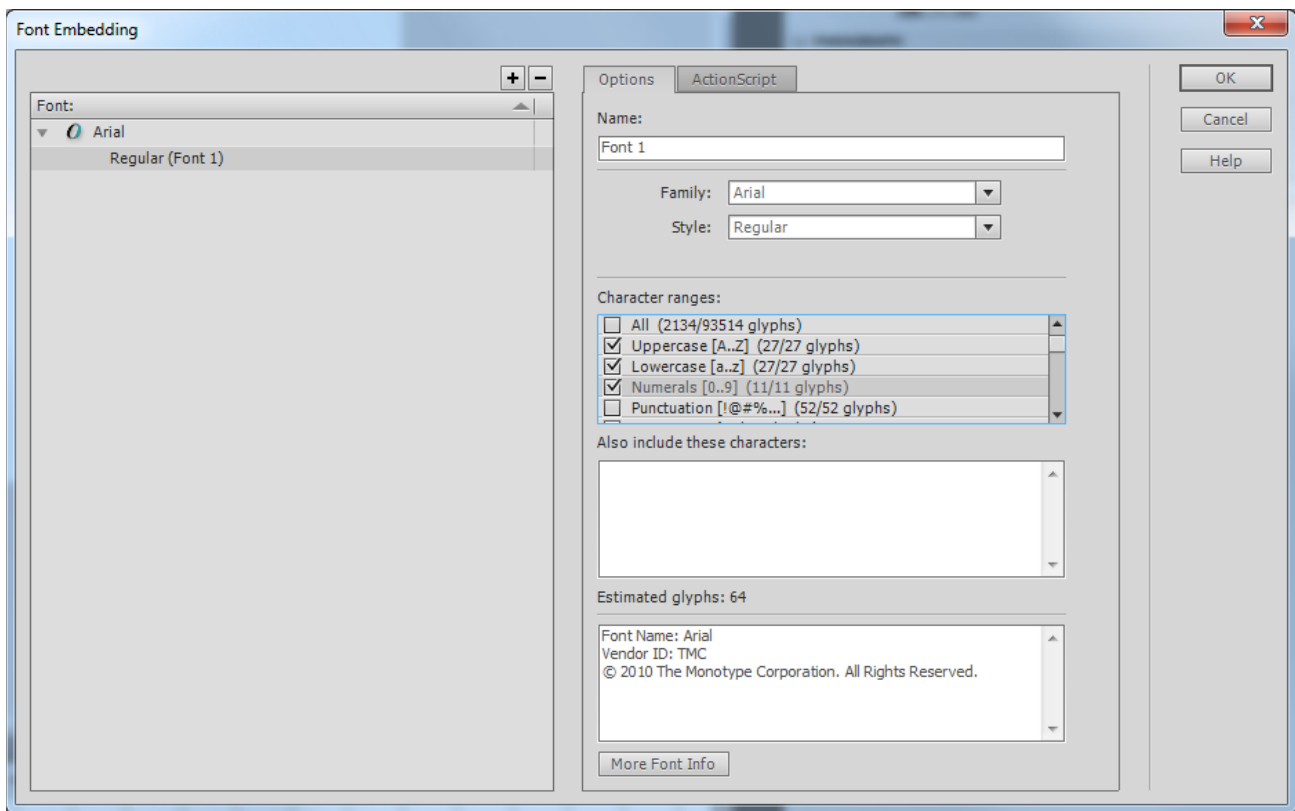
“ ” “ ”

1.1.2

TextField

Scaleform
Embed

Scaleform



“Style”

Arial

Arial

Flash Studio

“

”

SWF/GFx

Scaleform
Linux

Xbox 360 PS3 Wii

Windows Mac

1.1.3

Scaleform ,

		SWF	Scaleform
1		1 KB	450 K
114	– +	12 KB	480 K
596	–	70 KB	630 K
7,583	–	2,089 KB	3,500 K
18,437	–	5,131 KB	8,000 K

“Arial Unicode MS”

500 150K

1.1.4

1.

Flash

2.

Scaleform /

SWF/GFx

Scaleform

3.

IME

4.

GfxEport -fc

10%-30%

5. 3

Scaleform

IME

1.2

Scaleform

- 1 _____
- 2 _____
- 3 _____
- 4 -_____

Scaleform

Scaleform

2

- Scaleform
-
-

Flash

SWF/GFx

Scaleform
3 “ ”

Scaleform

Scaleform
FreeType 2

Scaleform
“ ” Scaleform
gfxexport

2 1

1.

2.

3.

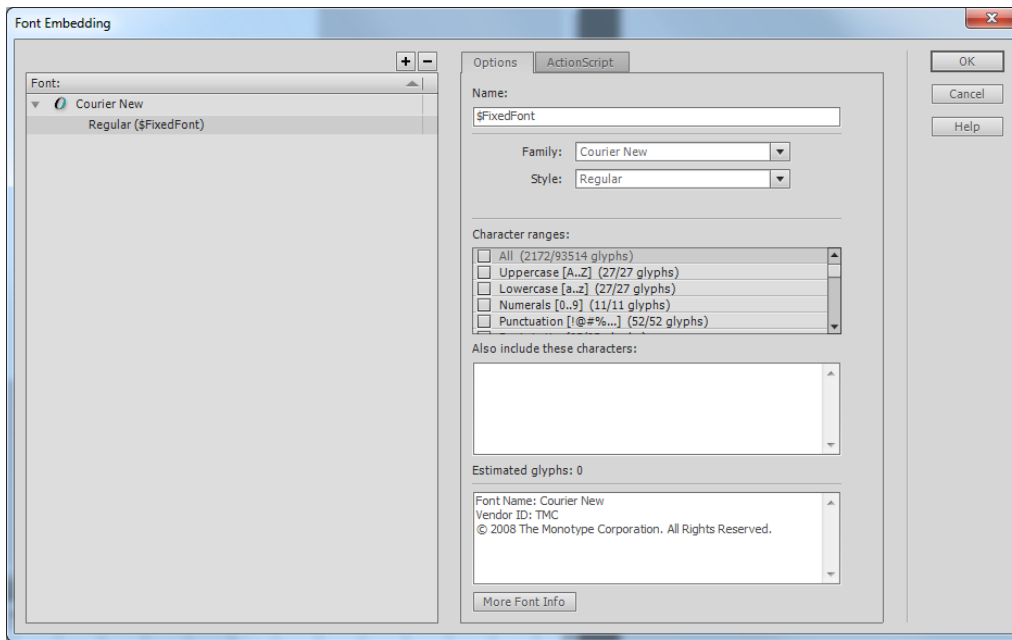
Scaleform

"fonts_en.swf" "fonts_kr.swf"
Flash

2.1

Flash Studio

"New Font..."



FLA

“\$FixedFont”

“\$ArialGame*”

“\$”

“ ”

“Font rendering method”

“

”

Flash

Studio

2.1.1

FLA

FLA

FLA

/

“Copy”

“Paste”

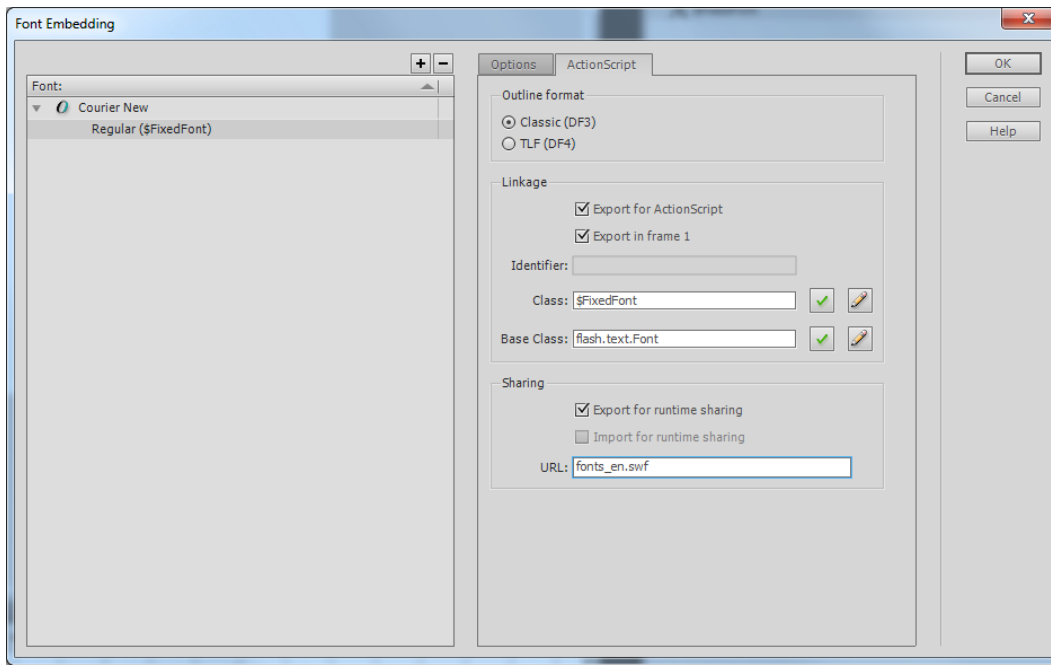
FLA

Flash

/

“Properties”

“ActionScript”



"OK"

Flash

"Export for ActionScript"

ActionScript

"Export for runtime sharing"

"Export in first frame"

URL

SWF

GfX::FontLib

URL

"fonts_en.swf"

GfX

```
Loader.SetDefaultFontLibName("fonts_en.swf");
```

fonconfig.txt GfXPlayer

```
fontlib "fonts_en.swf"
```

fontlib

"Export: \$identifier"

/

SWF

Scaleform

SWF

TextField.htmlText

TextFormat

“Arial”

2.1.2

CS4 Flash Studio

Windows	“	\	\	”	“	Unicode
”	“	”	243			
11920						‘gfxfontlib.swf’

Flash CS5	(Glyph)	'gfxfontlib.swf'
	'fonts_en.swf'	'fonts_jp.swf'

CS4 Flash Studio

'gfxfontlib.swf'

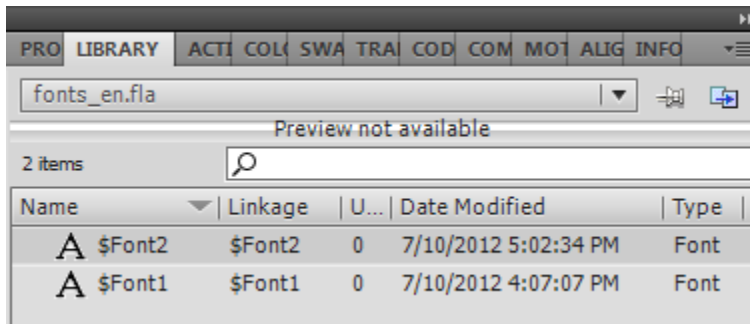
GFx 4.0

2.2 *gfxfontlib.swf*

Flash Studio FLA

1. `'$TitleFont'` `'$NormalFont'`
2. `FLA`
3. `1`
- 4.
- 5.
6. `fonts_<lang>.swf` `'lang'` `'en'` `'ru'`
`'jp'` `'cn'` `'kr'` `FontMap`

FLA



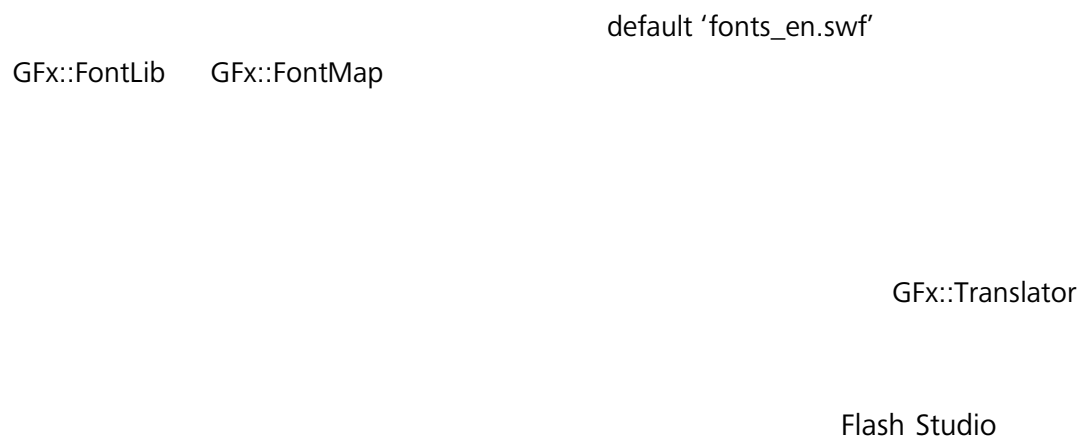
The screenshot shows the Adobe Flash Library panel. At the top, there is a tab bar with 'PRO' and 'LIBRARY' selected. Below the tabs is a search bar containing 'fonts_en fla'. A message 'Preview not available' is displayed. Below this, it says '2 items'. A table lists two font assets:

Name	Linkage	U...	Date Modified	Type
A \$Font2	\$Font2	0	7/10/2012 5:02:34 PM	Font
A \$Font1	\$Font1	0	7/10/2012 4:07:07 PM	Font

/

3 2

- 1.
- 2.
- 3.



3.1

Flash / 1

```
Loader.SetDefaultFontLibName("fonts_kr.swf");
```

SWF

CreateInstance

- 1.
- 2.
- 3.

“\$NormalFont”

Scaleform

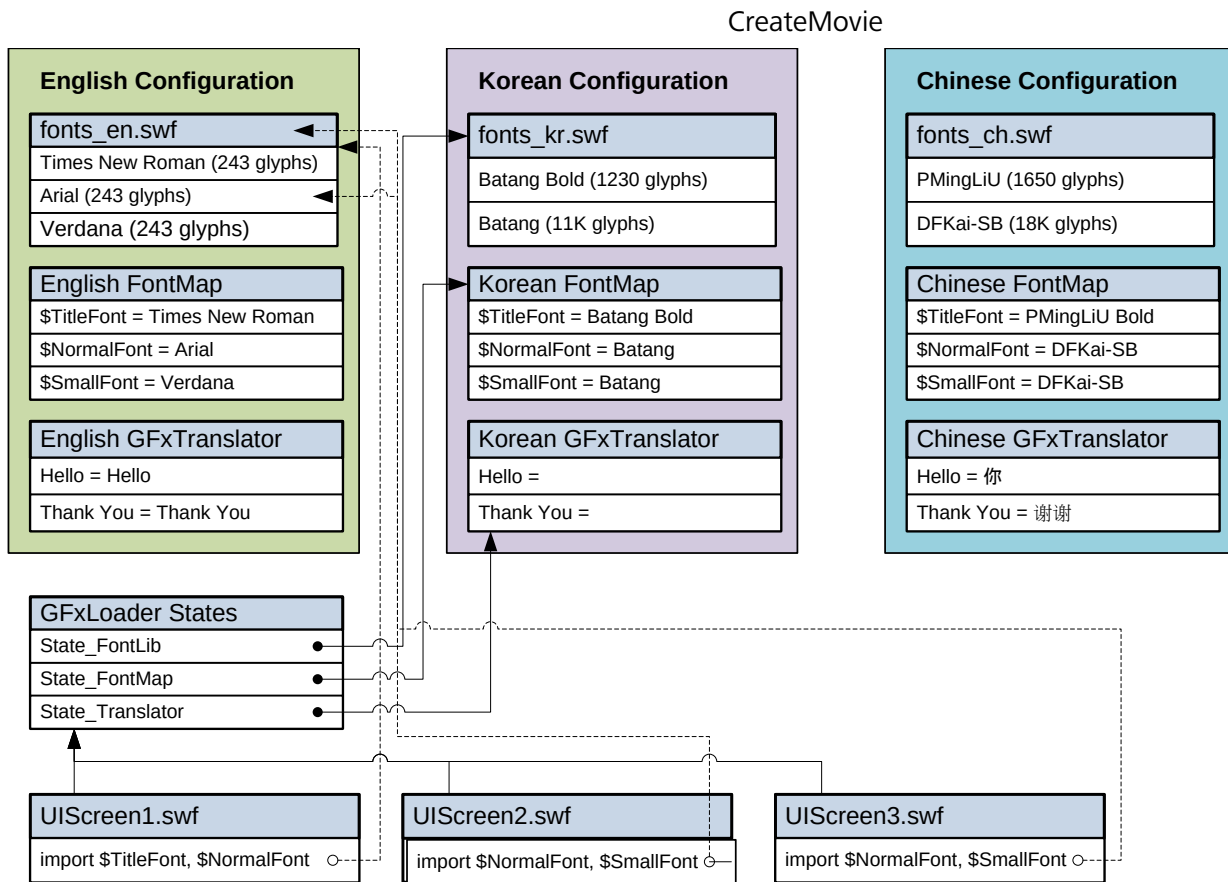
Gfx::Translator

Gfx::FontLib

Gfx::FontMap

Gfx::Loader

Gfx::Loader::CreateMovie



“UIScreen1.swf”

“UIScreen3.swf”

“fonts_en.swf”

(2) UI (1) Loader::SetDefaultFontLibName()
Gfx::FontLib
,
“fonts_kr.swf”

UIScreen

3.1.1

SWF Scaleform
Gfx::FontMap Gfx::Translator
Scaleform “fontconfig.txt”

1. 1 “fonts_en.swf”
2. Gfx::Translator
3. Gfx::FontMap
4. 为每个目标语言创建针对语言的字体文件，如 “fonts_kr.swf”。

1. Gfx::Loader Gfx::Translator
2. Gfx::FontMap Gfx::Loader MapFont
3 Gfx::FontMap
3. Gfx::FontProvider /
Gfx::FontPackParams GlyphCacheConfig
4. Gfx::FontLib Gfx::Loader

5. swf SetDefaultFontLibName(filename)

Loader.SetDefaultFontLibName("fonts_en.swf");

Gfx::FontLib SWF/GFx
Gfx::FontLib::AddFromFile

6. Gfx::Loader::CreateMovie

3.1.2 Scaleform Player

Scaleform Player "fontconfig.txt"

SWF/GFx 'fontconfig.txt'

/fc Ctrl+N F2

HUD

8 ASCII UTF-16

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$TitleFont" = "Times New Roman" Normal
map "$NormalFont" = "Arial " Normal
map "$SmallFont" = "Verdana" Normal
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
tr "Hello" = " "
tr "Thank You" = " "
```

```
[FontConfig "Chinese"]
fontlib "fonts_ch.swf"
map "$TitleFont" = "PMingLiU" Bold
map "$NormalFont" = "DFKai-SB" Normal
map "$SmallFont" = "DFKai-SB" Normal
tr "Hello" = " "
tr "Thank You" = " "
```

[FontConfig "name"]

Scaleform Player HUD

fontlib "fontfile"	SWF/GFxGFx::FontLibfontlib
map "\$UIFont" = "PMingLiU"	GFx::FontMapGFx::FontMap\$UIFont“fonts_en.swf”UI
tr "Hello" = " "	

Scaleform

XMLFontConfigParser.h/.cppScaleform

Bin\Data\AS2\Samples\FontConfig and Bin\Data\AS3\Samples\FontConfig

Scaleform“sample.swf”“fontconfig.txt”“

sample.swf”Scaleform Player“fontconfig.txt”

3.1.3

GFx::FontMap

“Arial” “Verdana”

Scaleform Player

“Arial”

GFx::FontLib

“font_kr.swf” “Batang” “Batang”

GFx::FontLibGFx::FontProviderWin32

GFx::FontProviderFT2

- FlashAdobe Flash Player
- ScaleformFlash

- “ Anti-alias for animation ”

ActionScript

TextField.antiAliasType

Scaleform Player
- 当使用设备字体时，开发人员必须重新编辑所有用户界面资源才能修改开发字体。当然，在 Scaleform 中可以使用字体映射，但当文件在 Adobe Flash 播放器中被测试时字体映射就不能使用了。相反，如果使用导入的字体替换，就可以通过简单地编辑和重新生成默认字体库文件 (“fonts_en.swf”) 来选择默认开发字体。
- ActionScript

TextFormat

Flash

“ Arial ”

“ \$TitleFont ”

“ fonts_en.swf ”

“ fonts_en.swf ”

3.1.4

- | | SWF | SWF | fontconfig.txt |
|-----|---|-------------------------------|-------------------------------------|
| 1. | FLA | ActionScript 2.0 | 3.0 'main-app.flaswf' |
| 2. | FLA | ActionScript | 'fonts_en.swf' |
| 3. | 'Library' | 'New Font...' | ... |
| 4. | \$Font1 "Family" | "Arial" "Style" | "Regular" English 'Basic |
| 5. | 'ActionScript' | 'Export for ActionScript' | ActionScript 'Export for |
| 6. | frame 1' 1 | 'Export for run-time sharing' | 'URL' |
| 7. | 'fonts_en.swf' | 'OK' | ActionScript 3.0 Flash Studio |
| 8. | 'A definition for this class could not be found...' | ... | — |
| 9. | 'OK' | 3 – 5 | '\$Font2' |
| 10. | 'Family' | 'Style' | 'Family' / 'Style' |
| 11. | | | Flash Studio |
| 12. | | | 'Times New Roman' |
| 13. | FLA | ActionScript | 'fonts_kr.flaswf' 'fonts_ru.flaswf' |
| 14. | | | Korean 'fonts_kr.flaswf' |
| 15. | 'Library' | 'New Font...' | ... |
| 16. | \$Font1 "Family" | "바탕" | "Style" |
| 17. | "Regular" | | |

9. "Character ranges" 'Korean Hangul
(All)'
10. 'ActionScript' 'Export for ActionScript' ActionScript 'Export for
frame 1' 1 'Export for run-time sharing' 'URL'
'fonts_en.swf' 'OK' ActionScript 3.0 Flash Studio
'A definition for this class could not be found...'
'OK'
11. 8 – 10 '\$Font2' 'Family'
'바탕체'
12. 'fonts_en.swf' 'fonts_kr.swf'
13. main-app.fl'a 'fonts_en.swf
Library '\$Font1' '\$Font2' Ctrl-C ->
'Copy'
14. 'main-app.fl'a Library Ctrl-V -> 'Paste'
'Import:'
15. 'main-app.fl'a 'Classic Text' 'Dynamic
Text' 'Family' '\$Font1*' '\$TEXT1'
ID
16. '\$Font2*' '\$TEXT2'
17. 'main-app.swf'
18. swf fontconfig.txt 'Notepad'
Unicode UTF-8

```
[FontConfig "English"]
fontlib "fonts_en.swf"
map "$Font1" = "Arial"
map "$Font2" = "Times New Roman"
tr "$TEXT1" = "This is"
tr "$TEXT2" = "ENGLISH!"
```

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$Font1" = "Batang"
map "$Font2" = "BatangChe"
tr "$TEXT1" = "이것"
tr "$TEXT2" = "은 한국이다!"
```

重要提示：

"Batang" "BatangChe"
SWF

"바탕" "바탕체" Flash SWF
fontconfig.txt
gfxexport -fntlst <swfname>

*.lst

19.	Unicode	UTF-8	GFxPlayer	'main-
app.swf'	'This is'	'English!'	Ctrl-N	
Korean	fonts_kr.swf		fonts_kr.swf	

3.2

			SWF/GFx	
	“UIScreen.flas”		SWF	“
UIScreen.swf”	“UIScreen.swf”			
		600K		
•				
•				
•				
•	IME			
			Flash Studio	“
”		Ctrl + F11		
“Replace strings”	“manually using stage language”		Flash	
Scaleform				

4 3

Flash

HTML

1.

2. SWF / GfX

3. GfX::FontLib

SWF/GfX

4. GfX::FontProviderWin32

Flash

GfX::FontLib GfX::FontMap GfX::FontProvider
GfX::FontLib GfX::FontMap SWF/GfX

GfX::Loader::SetFontProvider

SWF

SWF

Scaleform

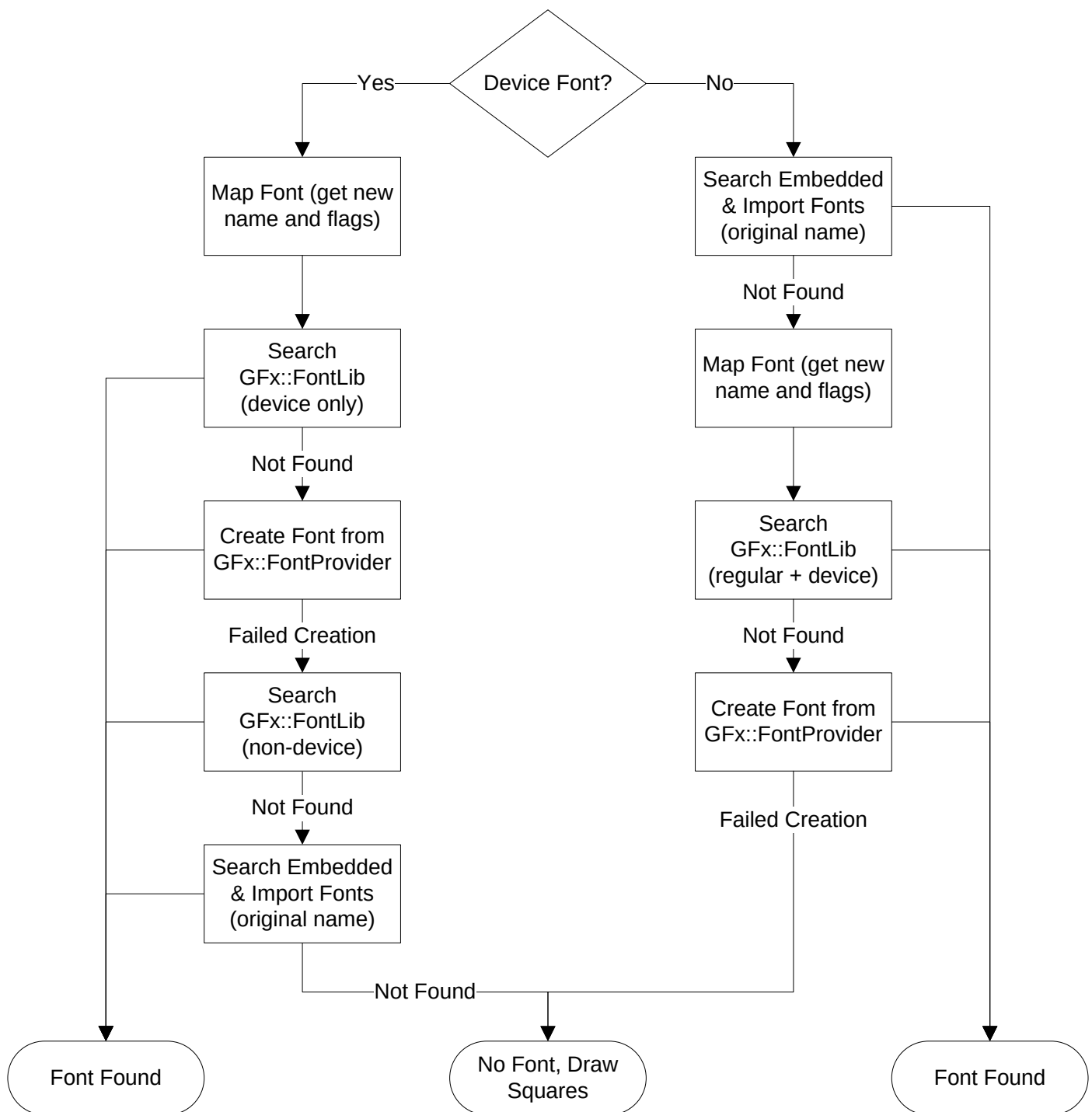
- GfX::FontProviderWin32 – Win32 APIs
- GfX::FontProviderFT2 – David Turner FreeType-2

4.1

Scaleform
“Use device fonts”

Adobe Flash

Scaleform



Gfx::FontLib Gfx::FontProvider

4.2 Gfx::FontMap

Gfx::FontMap
Gfx::FontMap

3

```
[FontConfig "Korean"]
fontlib "fonts_kr.swf"
map "$TitleFont" = "Batang" Bold
map "$NormalFont" = "Batang" Normal
map "$SmallFont" = "Batang" Normal
```

“\$TitleFont ”

C++

```
#include "Gfx/Gfx_FontLib.h"
. . .
Ptr<FontMap> pfontMap = *new FontMap;
Loader.SetFontMap(pfontMap);

pfontMap->MapFont("$TitleFont", "Batang", FontMap::MFF_Bold scaleFactor = 1.0f);
pfontMap->MapFont("$NormalFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
pfontMap->MapFont("$SmallFont", "Batang", FontMap::MFF_Normal scaleFactor = 1.0f);
```

```
;      “$NormalFont” “$SmallFont”
      MapFont
      MFF_Original
      scaleFactor
```

scaleFactor 1.0f

Gfx::FontMap

Gfx::FontMap

Gfx::Loader::CreateMovie

Gfx::MovieDef

4.4.1

Scaleform

font APIs (Gfx::FontProviderWin32 and

Gfx::FontProviderFT2)

```
Font::NativeHintingRange vectorRange;  
Font::NativeHintingRange rasterRange;  
unsigned maxVectorHintedSize;  
unsigned maxRasterHintedSize;
```

VectorRange RasterRange

Font::DontHint -

Font::HintCJK -

Font::HintAll -

RasterRange

VectorRange

maxVectorHintedSize

maxRasterHintedSize

SimSun

abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ

vectorRange=GfxFont::DontHint
rasterRange=GfxFont::DontHint

傷偽機播德倅借俾僥僦僭僦僦僦僦僦僦僦僦僦僦僦僦僦
僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦

abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ

vectorRange=GfxFont::HintAll
rasterRange=GfxFont::DontHint

傷偽機播德倅借俾僥僦僭僦僦僦僦僦僦僦僦僦僦僦僦
僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦

abcdefghijklmnopqrstuvwxyz
ABCDEFGHIJKLMNOPQRSTUVWXYZ

vectorRange=GfxFont::HintAll
rasterRange=GfxFont::HintCJK

傷偽機播德倅借俾僥僦僭僦僦僦僦僦僦僦僦僦僦僦僦
僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦僦

“Arial Unicode MS”

abcdefghijklmnopqrstuvwxyz
 ABCDEFGHIJKLMNOPQRSTUVWXYZ
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞
 僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞僞

Adobe Flash

Scaleform

Win32 FreeType

```

vectorRange = Font::DontHint;
rasterRange = Font::HintCJK;
maxVectorHintedSize = 24;
maxRasterHintedSize = 24;

```

4.4.2

SetHinting()

SetHintingAllFonts()

```
fontProvider->SetHintingAllFonts(. . .);
```

```
fontProvider->SetHinting(fontName, . . .);
```

SetHinting()

SetHintingAllFonts()

```

void SetHintingAllFonts(Font::NativeHintingRange vectorRange,
                        Font::NativeHintingRange rasterRange,
                        unsigned maxVectorHintedSize=24,
                        unsigned maxRasterHintedSize=24);

```

```

void SetHinting(const char* Name,
                Font::NativeHintingRange vectorRange,
                Font::NativeHintingRange rasterRange,
                unsigned maxVectorHintedSize=24,
                unsigned maxRasterHintedSize=24);

```

Name

UTF-8

4.5 GfX::FontProviderFT2

David Turner FreeType-2 library
 GfX::FontProviderWin32 *GfX::FontProviderFT2*

FreeType

Windows MSVC

```

freetype<ver>.lib -           DLL
freetype<ver>_D.lib -         DLL
freetype<ver>MT.lib -         DLL(   CRT)
freetype<ver>MT_D.lib -       DLL (   CRT)

    "<ver>"      FreeType          2.1.9      219

    freetype2/include  freetype2/objs
  
```

GfX::FontProviderFT2

```

#include "GfX/GfX_FontProviderFT2.h"
. . .
Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
<Map Font to Files or Memory>
Loader.SetFontProvider(fontProvider);
  
```

FontProviderFT2(FT_Library lib=0);

```

    FreeType                                     FreeType
          FreeType                               Scaleform
                                     (
FT_Done_FreeType)                               GfX::Loader
          FreeType
fontProvider->GetFT_Library()                     FT_Library
          FreeType                               GfX::FontProviderFT2
FreeType
  
```


4.5.1 FreeType

Win32 API

FreeType

Gfx::FontProviderFT2

```
void MapFontToFile(const char* fontName, unsigned fontFlags,
                  const char* fileName, unsigned faceIndex=0,
                  Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                  Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                  unsigned maxVectorHintedSize=24,
                  unsigned maxRasterHintedSize=24);

    "fontName"                "Times New Roman"        "filename"
    "C:\\WINDOWS\\Fonts\\times.ttf"    "fontFlags"        "Font::FF_Bold "
    "Font::FF_Italic "    "Font::FF_BoldItalic"    "Font::FF_BoldItalic"
    "Font::FF_Bold | Font::FF_Italic"    "faceIndex"        FT_New_Face
    0
;                                Win32
    FreeType
```

4.5.2

FreeType

```
void MapFontToMemory(const char* fontName, unsigned fontFlags,
                    const char* fontData, unsigned dataSize,
                    unsigned faceIndex=0,
                    Font::NativeHintingRange vectorHintingRange = Font::DontHint,
                    Font::NativeHintingRange rasterHintingRange = Font::HintCJK,
                    unsigned maxVectorHintedSize=24,
                    unsigned maxRasterHintedSize=24);
```

```
FILE* fd = fopen("C:\\WINDOWS\\Fonts\\times.ttf", "rb");
if (fd) {
    fseek(fd, 0, SEEK_END);
    unsigned size = ftell(fd);
    fseek(fd, 0, SEEK_SET);
    char* font = (char*)malloc(size);
    fread(font, size, 1, fd);
    fclose(fd);
}
```

```

        fontProvider->MapFontToMemory("Times New Roman", 0, font, size);
    }

```

Gfx::Loader

FreeType

Scaleform

Windows

FreeType

```

Ptr<FontProviderFT2> fontProvider = *new FontProviderFT2;
fontProvider->MapFontToFile("Times New Roman", 0,
    "C:\\WINDOWS\\Fonts\\times.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\timesbd.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\timesi.ttf");
fontProvider->MapFontToFile("Times New Roman", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\timesbi.ttf");

fontProvider->MapFontToFile("Arial", 0,
    "C:\\WINDOWS\\Fonts\\arial.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\arialbd.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\ariali.ttf");
fontProvider->MapFontToFile("Arial", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\arialbi.ttf");

fontProvider->MapFontToFile("Verdana", 0,
    "C:\\WINDOWS\\Fonts\\verdana.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Bold,
    "C:\\WINDOWS\\Fonts\\verdanab.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_Italic,
    "C:\\WINDOWS\\Fonts\\verdanai.ttf");
fontProvider->MapFontToFile("Verdana", Font::FF_BoldItalic,
    "C:\\WINDOWS\\Fonts\\verdanaz.ttf");
. . .
Loader.SetFontProvider(fontProvider);

```

MapFontToFile()

5 4

Scaleform

1.

/

2.

gfxexport

3.

Gfx::FontPackParams

4.

PC Xbox 360 PS3

Nintendo Wii

Gfx::FontProviderWin32 Gfx::FontProviderFT2

CPU

PSP PS2

gfxexport

CPU

Scaleform

“

”

5.1

5.2

10000

更多) 的时候。在这些情况下, 应该使用动态缓存机制。此外, 动态缓存可以提供更多功能, 如优化可读性, 大大提高字体的质量。

```
renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
renderer      Render::Renderer2D
```

HAL

LRU

```
Render::GlyphCacheParams gcparams;  
gcparams.TextureWidth    = 1024;  
gcparams.TextureHeight   = 1024;  
gcparams.MaxNumTextures  = 1;  
gcparams.MaxSlotHeight   = 48;  
gcparams.SlotPadding      = 2;  
gcparams.TexUpdWidth     = 256;  
gcparams.TexUpdHeight    = 512;
```

```
renderer->GetGlyphCacheConfig()->SetParams(gcparams);
```

TextureWidth, TextureHeight -

2

MaxNumTextures -

MaxSlotHeight -

SlotPadding -

- H, E, F, T, U, V, W, X, Z
- z, x, v, w, y

Scaleform

“Warning: Font 'Arial': No hinting chars (any of 'HEFTUVWXZ' and 'zxvwy'). Auto-Hinting is disabled.”

“ ‘Arial’: ('HEFTUVWXZ' and 'zxvwy') ”

Flash “Zz” “Basic Latin” “ ”

5.3 – gfxexport

gfxexport SWF GFX

DDS TGA –fc gfxexport

-fc	
-fcl <size>	25625625% Flash1024
-fcm	BooleanFontCompactor 10- 70% – (Hash) 12321664

5.4 - gfxexport

–fonts gfxexport

Scaleform Player Scaleform

Gfx

gfxexport

-

-

-

CPU

.

Gfx::ImageCreator

Render::Renderer

-strip_font_shapes	GfX
-fi <format>	<format> TGA8() TGA24 () TGA32 DDS8 (-i option) TGA TGA8 DDS A8

5.2

5.5

GfXExport

Basic Latin

Scaleform

```
Ptr<FontPackParams> packParams = *new FontPackParams();
Loader.SetFontPackParams(packParams);
```

GfXExport Scaleform “
” GfXExport

```
Loader.GetFontPackParams()->SetUseSeparateTextures(Bool flag);
Loader.GetFontPackParams()->SetGlyphCountLimit(int lim);
Loader.GetFontPackParams()->SetTextureConfig(fontPackConfig);
```

SetUseSeparateTextures()

SetGlyphCountLimit() 0

500

```
SetTextureConfig()  
    FontPackParams::TextureConfig fontPackConfig;  
    fontPackConfig.NominalSize    = 48;  
    fontPackConfig.PadPixels      = 3;  
    fontPackConfig.TextureWidth   = 1024;  
    fontPackConfig.TextureHeight  = 1024;
```

NominalSize -

“NominalSize”

NominalSize

“

”

PadPixels - 1

TextureWidth, TextureHeight - 2

1) ;

```
2) Ptr<FontPackParams> packParams = *new FontPackParams();  
   Loader.SetFontPackParams(packParams);  
   ...  
   renderer->GetGlyphCacheConfig()->SetParams(Render::GlyphCacheParams(0));
```

```
3) Ptr<FontPackParams> packParams = *new FontPackParams();  
   Loader.SetFontPackParams(packParams);  
   Loader.GetFontPackParams()->SetGlyphCountLimit(500);  
   500
```

5.6

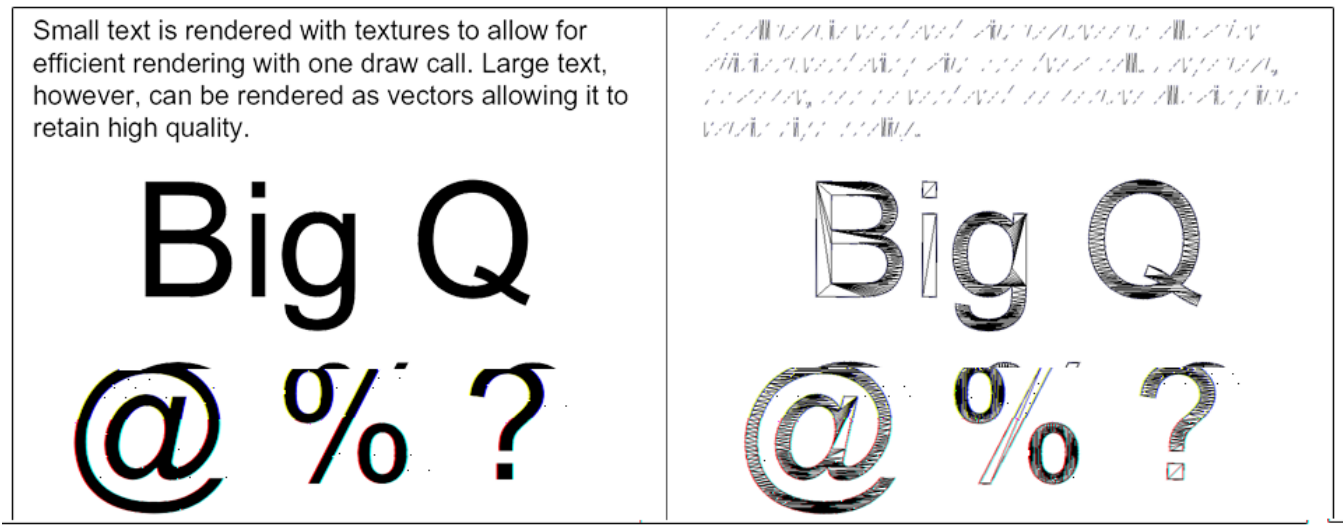
Flash

Scaleform

Scaleform

Scaleform Player

Ctrl+W



Render::GlyphCacheConfig::SetParams

MaxSlotHeight

SetMaxRasterScale

MaxRasterScale

1 1.25

GlyphCacheParams::MaxRasterScale

1.25

60

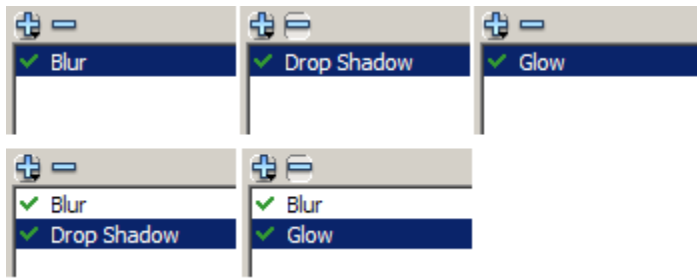
48

MaxRasterScale

gfxexport

GFx

-strip_font_shapes

[illegible]

Flash GfX

Text *Text*

“Knockout”

6.2

Flash “ ”

3x3 3x3

;

Flash Scaleform
Scaleform

“Quality: Low”

Scaleform
(sigma)

Scaleform

	Low quality	Medium quality	High quality
Flash	<i>Text</i>	<i>Text</i>	<i>Text</i>
GFx	<i>Text</i>	<i>Text</i>	<i>Text</i>

Flash

; Scaleform

Scaleform

<http://www.ph.tn.tudelft.nl/Courses/FIP/noframes/fip-Smoothin.html>.

6.3

Flash

Scaleform

Scaleform

/

6.4 *ActionScript*

Scaleform				DropShadowFilter	BlurFilter	ColorMatrixFilter	BevelFilter
	AS2	AS3	/				Flash